Predicting Low Income Status to Prioritize Voluntary Benefits Screening

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Executive Summary

Problem & Goal

A community nonprofit has limited capacity to provide one-on-one benefits screening to every applicant. This analysis develops a model to identify individuals most likely to earn $50,000 or less annually, helping staff prioritize voluntary invitations for income-tested benefits screening.

Data Description

The analysis used the UCI Adult Income dataset, containing 48,842 records, 14 predictor variables, and an income target. After cleaning, 48,813 records remained. The data were imbalanced: 76.06% of individuals were in the $50K-or-less category.

Data Cleaning

Duplicate records were removed, income labels were standardized, question-mark placeholders were converted to missing values, and a binary income target was created. For the random forest model, missing categorical values were imputed using the mode, nonpredictive variables were removed, and categorical variables were encoded.

Modeling

Three binary-classification models were evaluated: logistic regression, K-nearest neighbors, and random forest. Recall was prioritized because the organization benefits most from identifying as many potentially income-eligible individuals as possible.

Model Selection

Logistic regression was the strongest baseline model, with 84.86% accuracy and 93.06% recall. After tuning, the random forest model with a maximum tree depth of 15 was selected as the best overall option: it achieved 86.49% accuracy, 94.36% recall, 88.52% precision, and 62.19% specificity. It offered the best balance between reaching likely eligible clients and avoiding unnecessary outreach.

Managerial Insights

The nonprofit should use the selected model to prioritize—not determine—benefits-screening invitations. Lower education levels, fewer weekly work hours, and certain marital-status groups were associated with a higher likelihood of income at or below $50K; age alone was less informative. Final program eligibility should continue to be determined through the nonprofit's established screening criteria.

Problem & Goal

A community nonprofit offers income-tested programs, such as utility assistance, childcare subsidies, tax credits, and food assistance to the public. Public-benefit navigators within the organization can help only a fraction of applicants for its income-tested programs. At intake, staff capture demographic and employment information (age, education, occupation, marital status, work hours, etc.) but do not have a verified annual income figure, and proactively offering every client a full one-on-one screening during the initial visit is not feasible. As a result, public-benefit navigators lack a systematic way to determine who should apply for income-tested programs, so application help relies on staff intuition, which is hard to defend to funders and may miss the people most likely to benefit from screening. This project aims to build and compare classification models that predict whether an individual falls within the dataset's ≤$50K income category, then select the model that best helps the organization prioritize voluntary invitations for a full benefits-eligibility screening.

Data Description

This project uses the Census Income (Adult Income) dataset, originally sourced from the University of California, Irvine (UCI) Machine Learning Repository. The dataset is multivariate, is intended for use in the social sciences, and is suitable for classification tasks. The dataset contains 48,842 instances, 14 features (independent variables), and one target variable, income, which serves as the dependent variable for the classification task. To further describe the dataset, descriptive statistics and visualizations were generated using the cleaned dataset. The subsequent Data Cleaning section discusses the preprocessing procedures used to prepare the data.

Table 1

Dataset Variables

Variable Name	Type	Description
age	Integer	Age of the individual
workclass	Categorical	Work classification
fnlwgt	Integer	Assigned sampling weight
education	Categorical	Highest level of education
education-num	Integer	Numerical representation of education level
marital-status	Categorical	Marital status of the individual
occupation	Categorical	Type of occupation
relationship	Categorical	Individual's relationship within the household
race	Categorical	Race category of the individual
sex	Binary	Sex of the individual
capital-gain	Integer	Reported capital gains
capital-loss	Integer	Reported capital losses
hours-per-week	Integer	Number of hours worked per week
native-country	Categorical	Country of origin
income	Binary	Income classification: ≤$50K or >$50K

Note. The income variable is the target variable, classified as ≤$50K or >$50K annually. The remaining variables represent attributes available as predictors in the original Adult Income dataset.

The summary descriptive statistics show that age has a median of 37.0 and upper quartile of 48.0. At the same time, its mean is slightly higher than the median, and its maximum is considerably higher than both;; these statistics suggest a relatively balanced distribution around its center, with a slight right (positive) skew due to higher values pulling the mean upward. Education_level has a median of 10 and an upper quartile of 12.0. At the same time, its mean is approximately equal to the median, and its maximum is slightly higher than both; these statistics suggest a relatively balanced distribution around its center, with a very slight right (positive) skew due to higher values pulling the mean upward. Capital_gain has a median of 0 and an upper quartile of 0. At the same time, its mean and max are considerably higher than both statistics, suggesting a strong right (positive) skew driven by larger values pulling the mean upward. Capital_loss has a median of 0 and an upper quartile of 0. At the same time, its mean and max

are considerably higher than both statistics, suggesting a strong right (positive) skew driven by larger values pulling the mean upward. Hours_per_week has a median of 40 and an upper quartile of 45. At the same time, its mean is approximately equal to the median, and the max is considerably higher than both, suggesting a relatively balanced distribution around its center with a very slight right (positive) skew due to higher values pulling the mean upward. Overall, the summary statistics suggest skewness in capital_gain and capital_loss, and relative balance around their centers in education_num and hours_per_week, with a very slight right (positive) skew.

Table 2

Summary Statistics

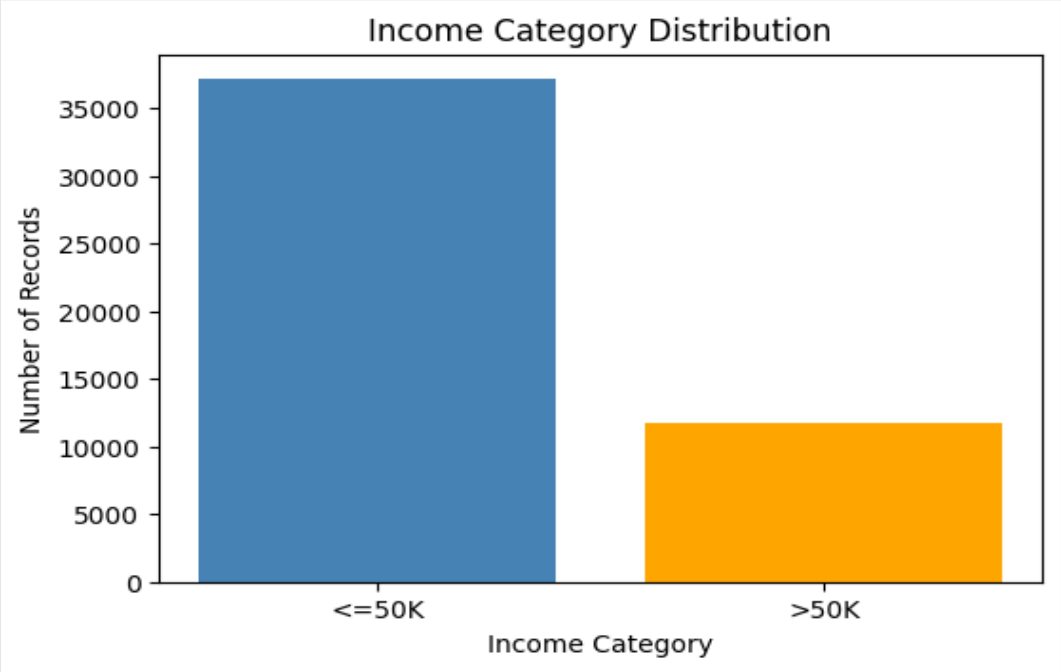
	count	mean	std	min	25%	50%	75%	max	median	IQR
age	48813.0	38.647348	13.709005	17.0	28.0	37.0	48.0	90.0	37.0	20.0
education_num	48813.0	10.078688	2.570257	1.0	9.0	10.0	12.0	16.0	10.0	3.0
capital_gain	48813.0	1079.708705	7454.185982	0.0	0.0	0.0	0.0	99999.0	0.0	0.0
capital_loss	48813.0	87.554299	403.118605	0.0	0.0	0.0	0.0	4356.0	0.0	0.0
hours_per_week	48813.0	40.425051	12.390954	1.0	40.0	40.0	45.0	99.0	40.0	5.0

Note. Summary statistics are based on the cleaned dataset (n = 48,813). SD = standard deviation; IQR = interquartile range. The 25%, 50%, and 75% columns represent the first quartile (Q1), median (Q2), and third quartile (Q3), respectively.

To further explore the dataset and identify meaningful patterns, visualizations were used to examine the distribution of the target variable and its relationships with selected categorical and numerical variables. The income category distribution shows that the ≤$50K category [37,128] ([76.06]%) contains substantially more observations than the >$50K category [11,685]

([23.93]%). This indicates that the target variable is imbalanced, with most observations in the ≤$50K category.

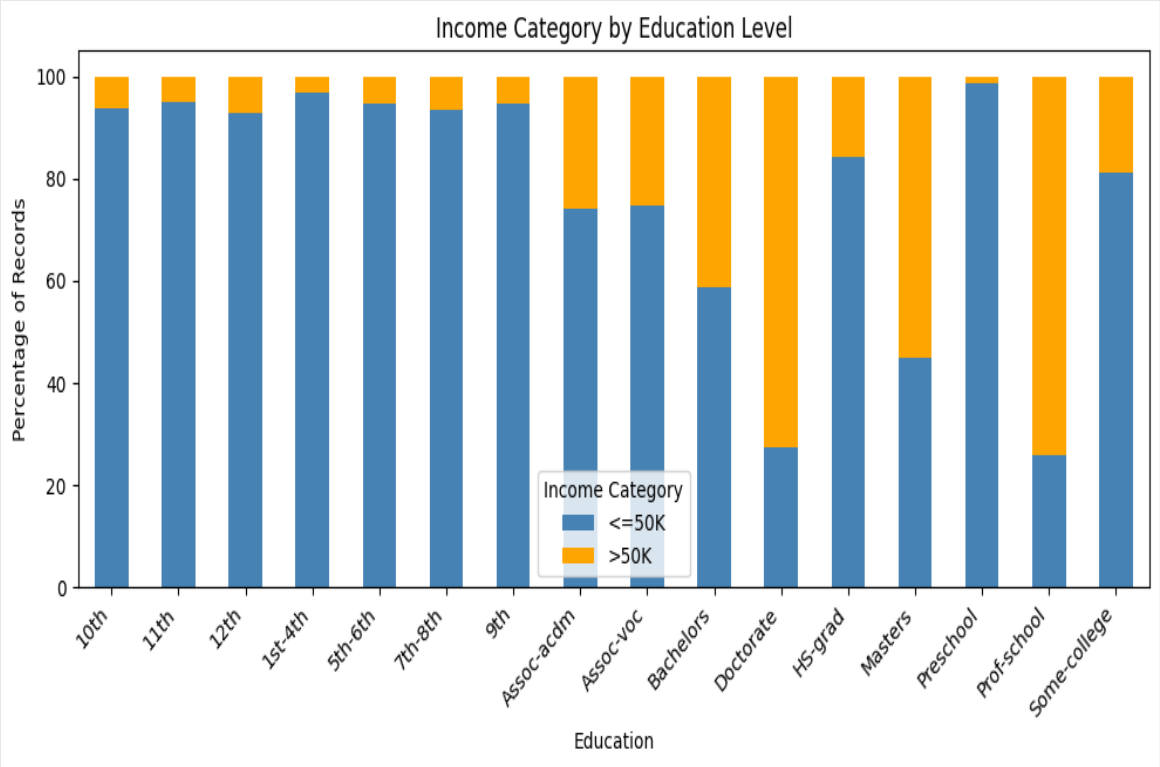
Figure 1



Note. The figure presents the number of individuals within each income category in the cleaned dataset. Individuals earning ≤$50K represent the majority of observations compared with individuals earning >$50K.

The income category by education level shows that lower education levels have substantially higher proportions of individuals earning ≤$50K, while advanced education levels have higher proportions earning >$50K. For example, about 70% of individuals with a doctorate earn >$50K, compared with a small proportion in lower education categories.

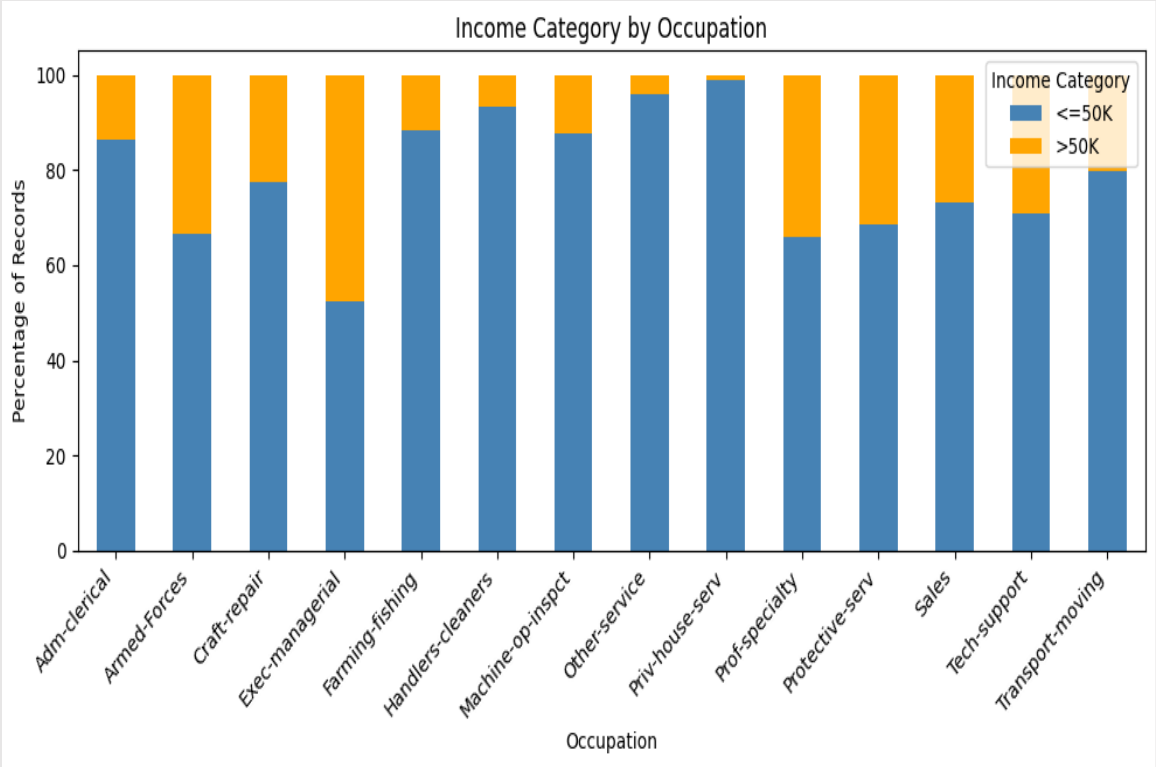
Figure 2



Note. The figure presents the percentage distribution of income categories within each education level. Blue represents individuals earning ≤$50K annually, and orange represents individuals earning >$50K annually. Percentages within each education category sum to 100%.

The income category by occupation shows that income levels vary within occupations, with most occupations having a higher proportion of individuals earning ≤$50K. Individuals within the same occupation may hold different positions or seniority levels that contribute to income differences. Therefore, individuals within these occupations who earn ≤$50K may be more likely to meet the income requirements for income-tested programs offered by the community nonprofit.

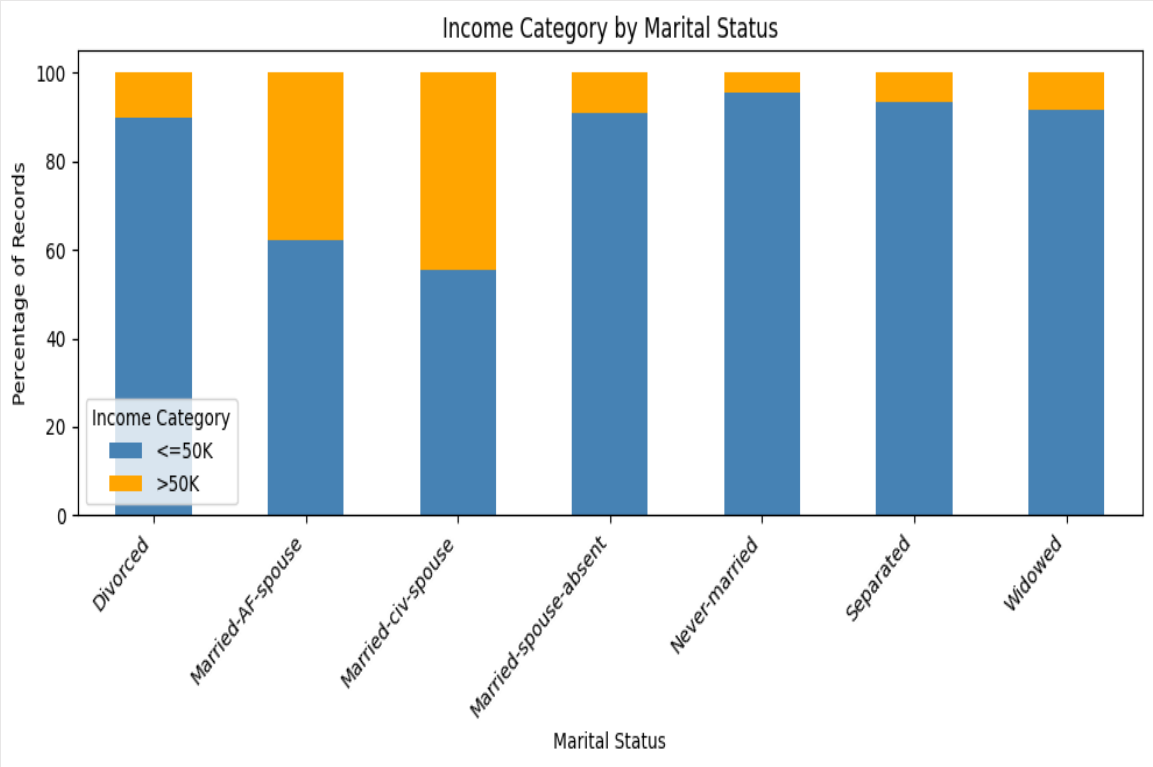
Figure 3



Note. The figure presents the percentage distribution of income categories within each occupation category. Blue represents individuals earning ≤$50K annually, and orange represents individuals earning >$50K annually. Percentages within each occupation category sum to 100%.

The income category by marital status shows that individuals earning ≤$50K were more highly represented across all marital-status categories. However, representation varied, with a more even distribution among married individuals; those earning ≤$50K remained the majority regardless of marital status. This suggests that marital status alone may not strongly distinguish individuals who are more likely to meet the income requirements for income-tested programs offered by the community nonprofit.

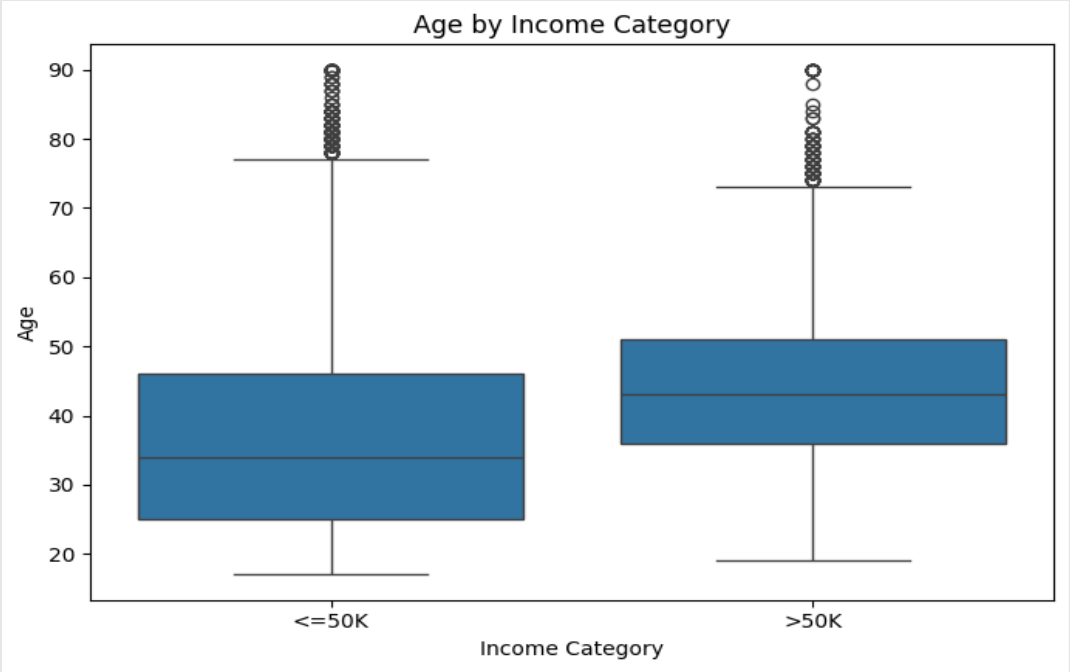
Figure 4



Note. The figure presents the percentage distribution of income categories within each marital-status category. Blue represents individuals earning ≤$50K annually, and orange represents individuals earning >$50K annually. Percentages within each marital-status category sum to 100%.

Individuals earning ≤$50K exhibited a wider spread in age than those earning >$50K, suggesting that the lower-income category is represented across a broader range of ages. Individuals earning >$50K had a higher median age; however, the distributions overlap substantially, suggesting that age alone may not strongly determine income category.

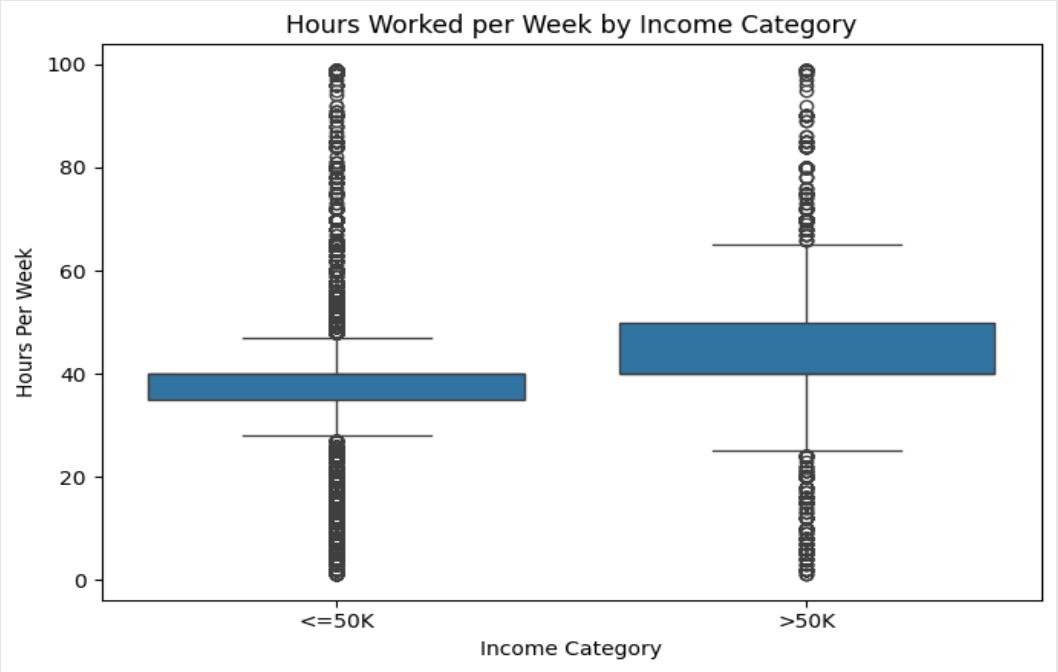
Figure 5



Note. The figure presents the distribution of age within each income category. The boxes represent the interquartile range (IQR), the horizontal line within each box represents the median, the whiskers represent values within 1.5 times the IQR, and individual points beyond the whiskers represent potential outliers.

Hours worked per week varied across income categories. Individuals earning ≤$50K had a median of about 35 hours per week, while those earning >$50K had a median of about 45 hours. Both income categories had numerous outliers above and below their typical ranges, with the ≤$50K group showing many observations with substantially fewer weekly hours. Overall, the distributions suggest some association between hours worked and income category, although both groups show considerable variation.

Figure 6



Note. The figure presents the distribution of hours worked per week within each income category. The boxes represent the interquartile range (IQR), the horizontal line within each box represents the median, the whiskers represent values within 1.5 times the IQR, and individual points beyond the whiskers represent potential outliers.

Data Cleaning

Baseline data cleaning was conducted to create a consistent dataset for use across all three predictive models before conducting individualized data cleaning for each model. The combined Adult Income dataset was first examined for duplicate records and identified duplicates were removed. The income target variable was standardized by removing trailing periods from category labels to ensure consistent values across the combined data. Missing-value indicators represented by question marks (?) were identified and converted to NaN so that missing observations were properly recognized. The binary target variable, income_binary, was then created, where 1 represented income at or below $50K and 0 represented income above $50K. Finally, a unique record ID was assigned to each observation to support consistent record

tracking across the modeling process. The resulting dataset was saved as the shared baseline dataset for subsequent model-specific preprocessing.

Following the baseline cleaning, each model underwent additional model-specific preprocessing based on its respective modeling requirements. Regarding the random forest model, remaining missing values in workclass, occupation, and native_country were addressed using mode imputation. Variables not selected as predictors – fnlwgt, education, race, sex, income, and record_id were removed from the modeling dataset. Numerical variables were examined for potential outliers; however, these observations were retained. The remaining categorical predictors were converted to categorical data types and numerically encoded using category codes to prepare the data for random forest modeling.

Modeling

The predictive models used to determine prioritization of individuals for voluntary invitations for full benefits-eligibility screening include logistic regression, KNN, and random forest. Each model is suitable for this particular analysis because the business problem involves a binary classification task, in which individuals are classified based on whether their income is ≤$50K or >$50K.

Table 3

Comparison of Predictive Classification Models

Model	Function	Advantages	Disadvantages	Dataset Suitability
Logistic Regression	Predicts probability of a binary outcome	Interpretable; provides probabilities	Limited with highly nonlinear relationships	Strong baseline for binary classification
KNN	Classifies using nearby observations	Flexible; captures nonlinear patterns	Sensitive to *k*, scaling, and dimensionality	Useful when similar observations share outcomes
Random Forest	Combines predictions from multiple decision trees	Captures complex relationships; reduces variance	Less interpretable; can overfit if unrestricted	Useful for complex predictor–target relationships

Note. Model characteristics and suitability were synthesized from the scikit-learn documentation for logistic regression, nearest neighbors, and random forests (Scikit-learn Developers, n.d.).

Model Selection

At baseline, logistic regression achieved the highest accuracy (84.86%) and recall (93.06%). Random forest achieved the highest precision (88.37%) and specificity (62.65%), while KNN produced the lowest overall baseline performance. Recall was designated as the primary evaluation metric; therefore, when comparing all models, logistic regression was selected as the strongest baseline model.

Table 4

Baseline Predictive Model Performance Comparison

Baseline Model	Accuracy	Recall	Specificity
Logistic Regression	84.86%	93.06%	59.55%
KNN	81.90%	89.60%	58.12%
Random Forest	84.73%	91.88%	62.65%

Note. All three models are compared using their baseline configurations prior to model-specific hyperparameter tuning.

Although the unrestricted random forest model achieved high training accuracy, the substantial gap between training and testing accuracy indicated overfitting, suggesting that the model did not generalize as well to unseen data. To address this issue, the model was fine-tuned by applying the max_depth hyperparameter to reduce tree depth and model complexity. Restricting tree depth improved test performance and reduced the overfitting observed in the unrestricted model. The fine-tuned configurations maintained high recall, ranging from 93.57% to 96.12%. A maximum depth of 15 achieved 94.36% recall while also producing the highest test accuracy (86.49%) and a specificity of 62.19%; compared with depth 10, depth 15 reduced recall slightly from 95.46% to 94.36%, but increased specificity from 58.21% to 62.19% and increased overall test accuracy from 86.35% to 86.49%.

Table 5

Random Forest Model Performance by Maximum Tree Depth

Estimators	Max Depth	Random State	Train Accuracy	Test Accuracy
500	Unrestricted	42	0.9704	0.8473
500	5	42	0.8542	0.8523
500	10	42	0.8702	0.8635
500	15	42	0.9008	0.8649
500	20	42	0.9378	0.8610

Note. All Random Forest configurations used 500 estimators and a random state of 42. The unrestricted model was trained without a specified maximum tree depth. Train and test accuracy represent the respective proportion of correctly classified observations in the training and testing datasets.

Table 6

Random Forest Classification Performance by Maximum Tree Depth

Max Depth	Accuracy	Precision	Recall	Specificity
Unrestricted	84.73%	88.37%	91.88%	62.65%
5	85.23%	85.98%	96.12%	51.59%
10	86.35%	87.58%	95.46%	58.21%
15	86.49%	88.52%	94.36%	62.19%
20	86.10%	88.66%	93.57%	63.02%

Note. Classification metrics were calculated using the test dataset. Accuracy represents overall classification performance, precision represents the proportion of predicted ≤$50K observations that were correctly classified, recall (sensitivity) represents the proportion of actual ≤$50K observations correctly identified, and specificity represents the proportion of actual >$50K observations correctly identified. The ≤$50K income category represents the positive class.

A maximum depth of 15 was selected as the final random forest configuration because it maintained high recall while providing a stronger overall balance between identifying individuals earning ≤$50K and correctly identifying those earning >$50K. KNN was also fine-tuned by evaluating alternative values of k, with k = 20 producing an accuracy of 83.80% and recall of 92.19%. Although logistic regression was selected based on the common baseline comparison, the fine-tuned random forest at a maximum depth of 15 outperformed both the fine-tuned KNN model and the reported baseline logistic regression model across accuracy, recall, and specificity. However, this comparison reflects baseline versus fine-tuned performance because the logistic regression model did not undergo fine-tuning.

Table 7

Baseline and Fine-Tuned Model Performance Comparison

Model / Configuration	Accuracy	Precision	Recall	Specificity
LR (Baseline)	84.86%	87.66%	93.06%	59.55%
KNN (Baseline)	81.90%	86.86%	89.60%	58.12%
KNN (Tuned (k=20))	83.80%	87.11%	92.19%	57.87%
RF (Baseline)	84.73%	88.37%	91.88%	62.65%
RF (Tuned (max_depth=15))	86.49%	88.52%	94.36%	62.19%

Note. Logistic Regression (LR); K-Nearest Neighbors (KNN); Random Forest (RF). Baseline results represent models prior to fine-tuning, while tuned results reflect the selected hyperparameter configurations.

Managerial Insights

Education level may help the nonprofit prioritize individuals for voluntary benefits-screening invitations because individuals with lower education levels had substantially higher observed proportions of income ≤$50K. Marital status may help the nonprofit prioritize

individuals for voluntary benefits-screening invitations because widowed, separated, never married, or divorced individuals may be more likely to fall within the ≤$50K income category than married individuals. Age alone may offer limited insight into which individuals are more likely to meet the income requirements for income-tested programs. Individuals earning ≤$50K who also work substantially fewer weekly hours may be a useful population to help the nonprofit prioritize individuals for voluntary benefits screening. Overall,, these findings suggest prioritizing the aforementioned populations for voluntary benefits-screening invitations when capacity is limited. However, the nonprofit should determine actual program eligibility using its established eligibility criteria rather than any one of these factors alone.

References

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